



Enhancing Travel Demand Modeling with Real-World Commercial Vehicle Data

Leveraging Altitude for Advanced Travel Demand Modeling in Ohio



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Geotab Corporate Overview



25+ YEARS

driving transformation



700+

Partners globally



5+ MILLION

Connected vehicles



2200+ EMPLOYEES

in 30+ countries



55,000+ CUSTOMERS

in 165 countries



>80B DATA POINTS

processed daily



Geotab ranked

#1 Telematics Provider

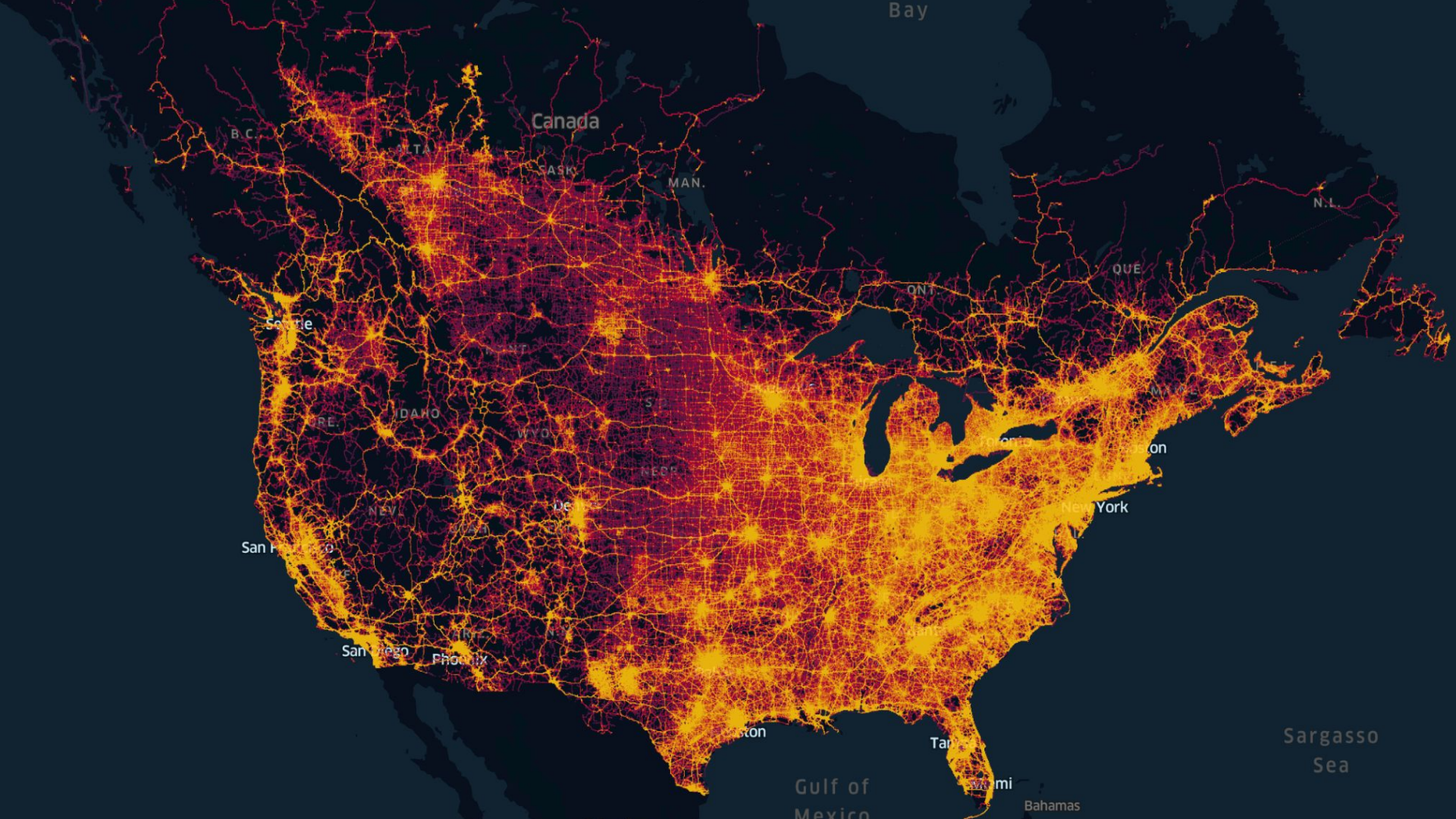
in the world by ABI Research



Unlocking Commercial Mobility Insights Through Data from 55,000 Fleets

- GPS Location
- Vehicle Health Monitoring
- Vehicle identification
- Real-time tracking
- Speed measurement
- Driving Behavior
- Trip and Routing information
- Travel speed and time
- Compliance (HOS monitoring)





Why Altitude ?

Privacy by design,
Altitude's **PII-safe layer**
ensures complete data
confidentiality.

Confidentially aggregated

Our stringent confidentiality and privacy standards allow us to deliver insights **without compromising driver, vehicle or company information.**

Insight integrity

As the **source & processor** of mobility data from over **4.7 million commercial vehicles and 55K fleets** we provide comprehensive insights with greater context.

Comprehensive History

Leveraging **extensive historical data** on vehicle movements supports long-term planning and decision-making, promoting effective analysis of travel patterns over time.

Added context

Beyond just traffic counts, we provide detailed traffic insights by **vehicle type** and **class, vocation, fuel type** and **industry** for granular road segments.

Advanced processing

Quickly process mobility data to generate visualizations tailored to any scenario, providing your team with actionable insights.

Complete Coverage

Altitude platform provides mobility insights for **99.9% of primary roadways** across North America.

How Can Altitude Support Freight and Travel Demand Modeling?

Observed Commercial Mobility Insights!

Real-World Movement Data

Altitude captures over **80 billion data points daily** from connected commercial vehicles across the U.S. and Canada, offering unmatched visibility into how freight and service vehicles move through your region.

Improved Model Calibration & Validation

Validate freight flows, stop behavior, and corridor usage with **observed first-party commercial vehicle data** to strengthen travel demand models.

Origin-Destination & Corridor Analysis

Identify top OD pairs, route choices, and stop clusters to inform model inputs and scenario planning.

Time-of-Day and Vehicle Class Patterns

Access insights by time period, day of week, and vehicle class (e.g., medium-duty, heavy-duty) to support dynamic modeling needs.

Customizable Regional Focus

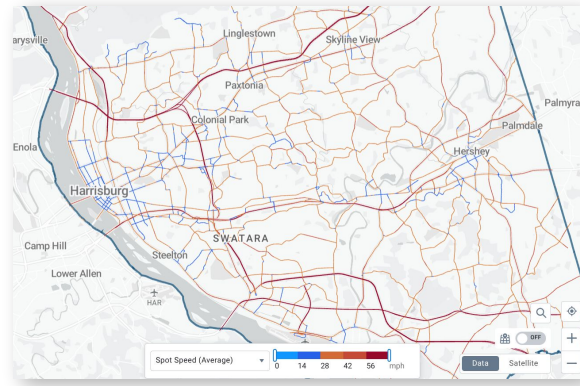
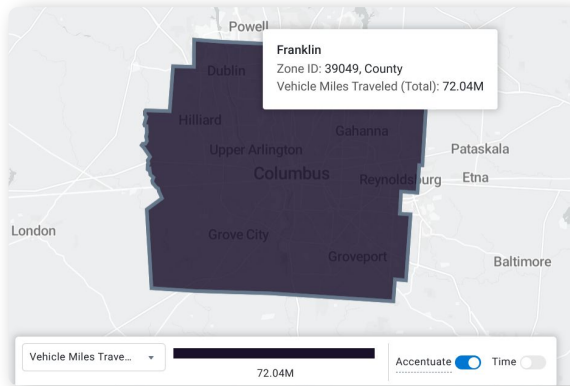
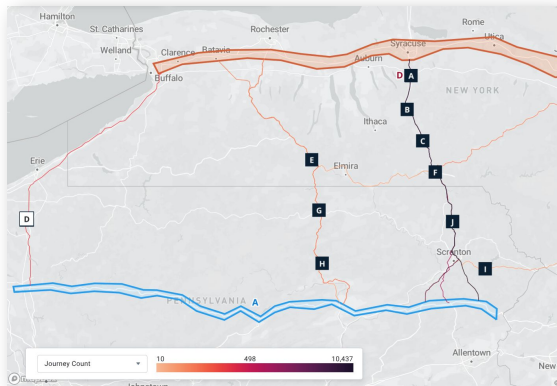
Filter and analyze vehicle movement data at the local, corridor, or regional level to align with your specific planning boundaries and modeling zones.

Data That Complements Your Models

Use Altitude data to fill gaps in traditional survey or count data, improving the accuracy of forecasts, especially for freight and commercial activity.

Altitude Modules & Features

Statewide travel demand insights at your fingertips



Origin & Destination

Analyze travel patterns between geographic zones and along key route connections between common origins and destinations.

Watch ☐



Regional Travel Metrics

Examine travel patterns with segmentable traffic counts, distance and idling metrics.

Read ☐

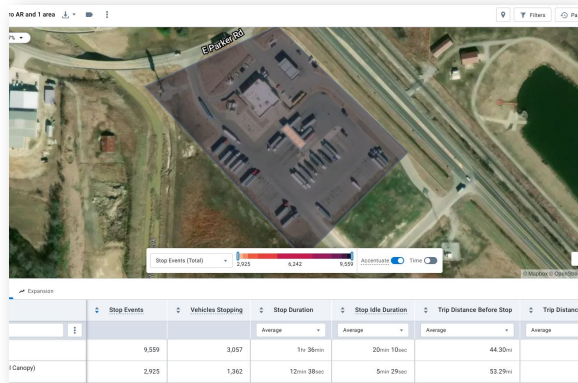


Traffic Analytics

Optimize traffic and safety investments informed by vehicle harsh braking events, travel time and speed metrics.

Watch ☐

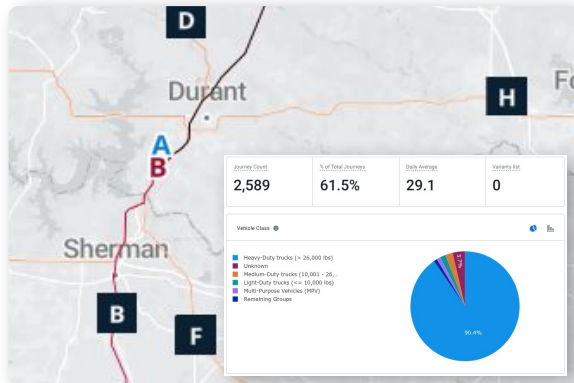
Your path to better planning



Stop Analytics Module

Identify where vehicles stop to plan parking, charging, and refueling.

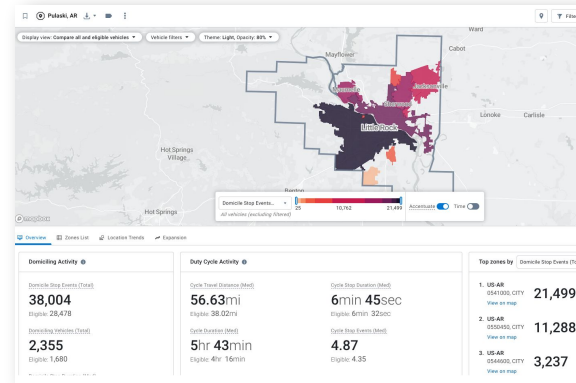
Watch [▶](#)



Route Analytics

Analyze routes, journey counts, and travel times to guide site selection planning.

[Read \[▶\]\(#\)](#)



Regional Domicile Analytics

Understand vehicle dwell times to optimize infrastructure.

Watch [▶](#)

Insight to Impact

Proven expertise in empowering data-driven decisions for smarter, safer, and more sustainable transportation.

- Trusted by leading agencies like Georgia DOT, Caltrans and New York State DOT.
- Delivered tangible results for NYC DOT and SCAG, improving planning and efficiency.
- Driving positive impact in sustainability, enhancing road safety through better transportation planning. **Supporting FHWA led safety initiatives.**
- Altitude is available through Sourcewell and The Eastern Transportation Coalition's Transportation Data Marketplace, giving public agencies a streamlined and trusted path for procurement.



Altitude → Travel Demand Modeling



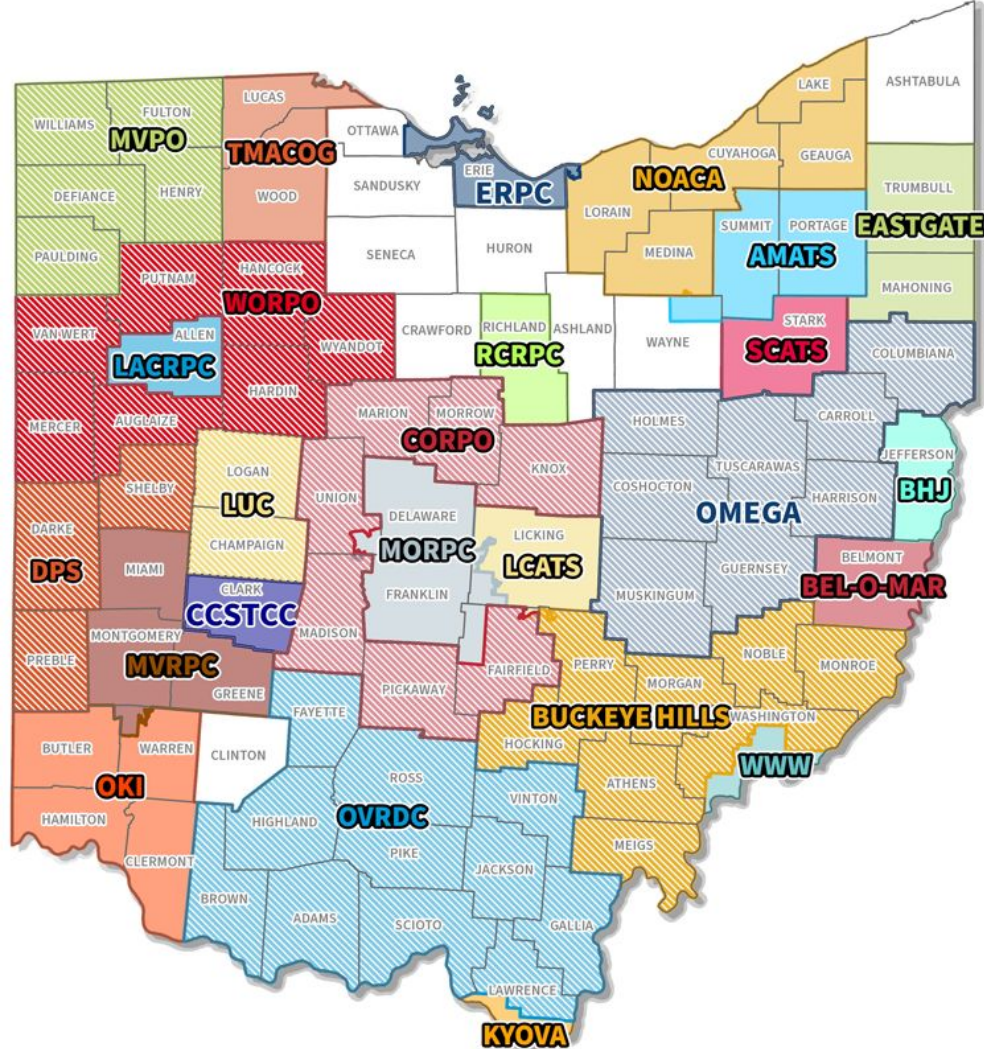
Altitude for Travel Demand Modeling

How can Altitude fit into the Travel Demand Model framework in Ohio?

Altitude modules can support trip-based and activity-based modeling.

- Trip Demand Generation (trip starts and ends)
- Origin-Destination (E-E, I-I, E-I, I-E)
- O-D Route Analysis
- Select Link Analysis (travel speeds, volumes)

Altitude can augment Base Model training with commercially-focused data.



Altitude Advantages

Commercial Vehicle data with Context

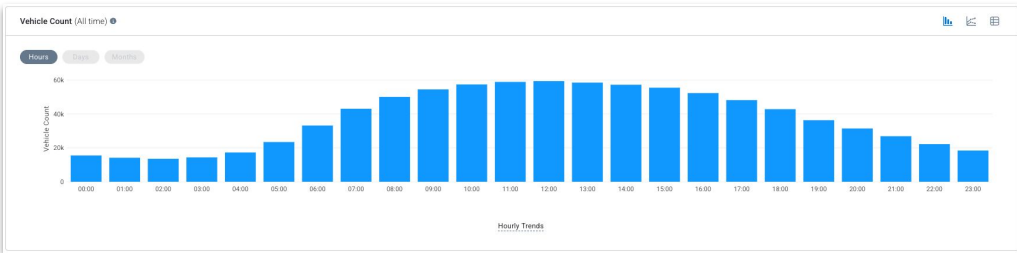
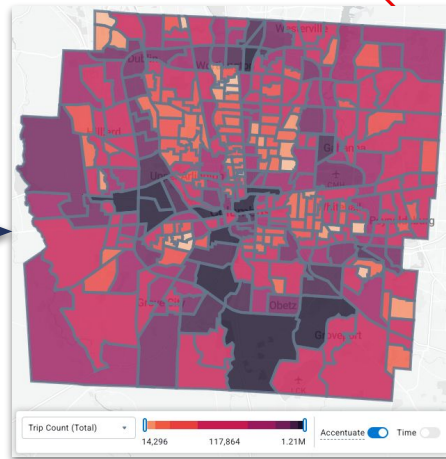
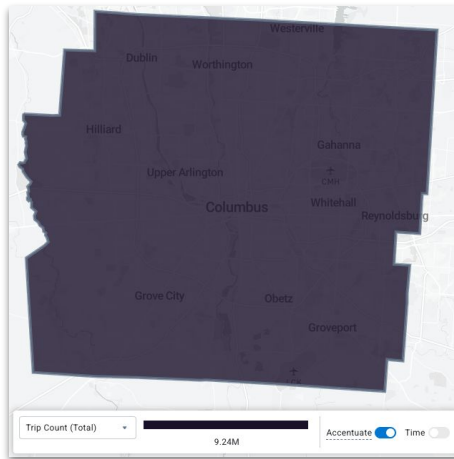
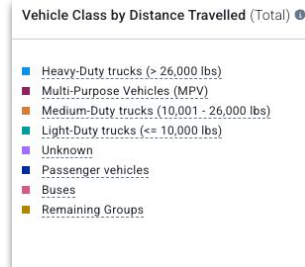
- Vocation: Long-Distance vs. Door-to-Door
- Vehicle classifications
- Freight demand generators by Industry

Analysis Granularity

- Spatial granularity
 - Standard & custom zones
- Temporal granularity

More Advantages

- Primary source of data
- 1 in 8 commercially registered vehicles in North American (US & Canada) data panel
- Expansion Factors provided



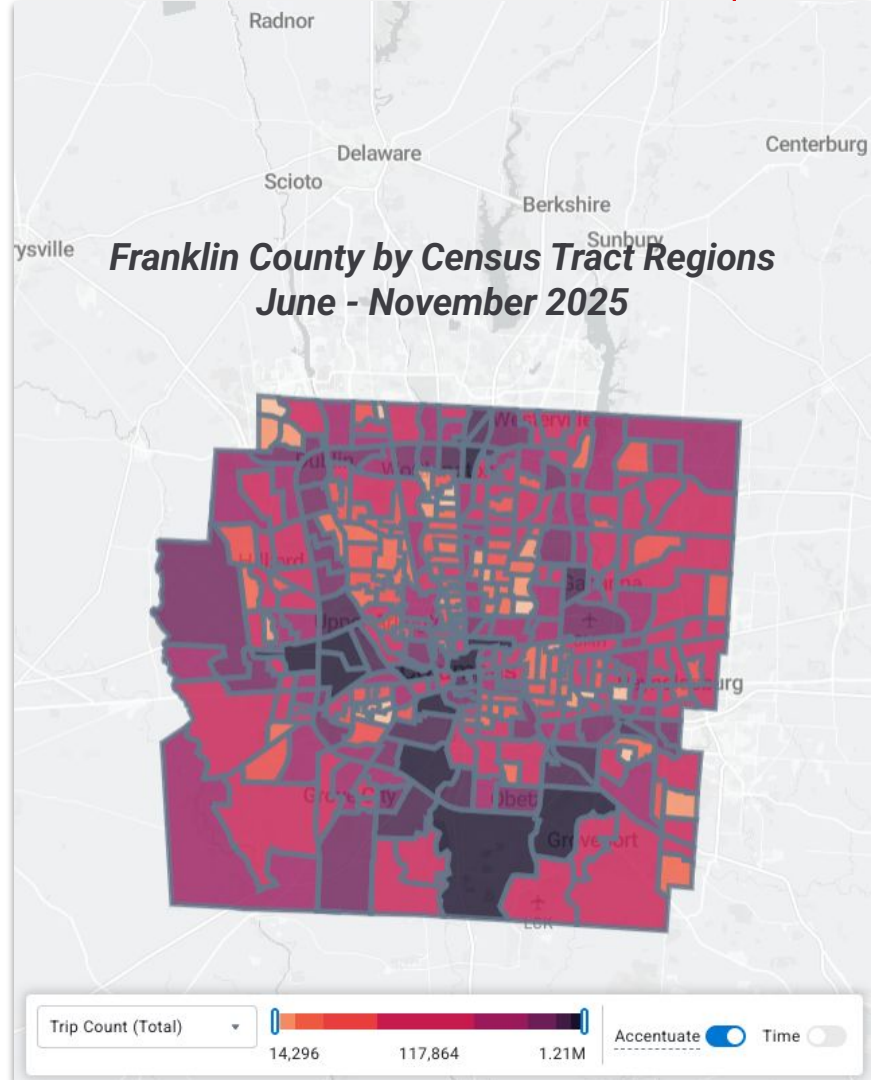
Trip Generation - Aggregated

- Altitude can provide trip counts, trip starts, and trip ends for standard zones (county, Census Tracts) and custom zones (TAZ).

Travel Distance and Activity ⓘ				
<u>Vehicle Miles Traveled (Total)</u>	<u>Vehicle Count (Total)</u>	<u>Trip Count (Total)</u>	<u>Trip Starts</u>	<u>Trip Ends</u>
76.07M	121,814	9.24M	13.6M	13.6M

- Metrics can be also be broken down by road types.

Road Type	Total VMT	Total Vehicle Count	Total Trip Count
Highways	45.54M	120,543	3.78M
Arterials	18.04M	264,050	13.61M
Collectors	6.19M	166,766	4.21M
Local Roads	1.18M	38,065	3.97M
Others	1.09M	61,884	2.88M

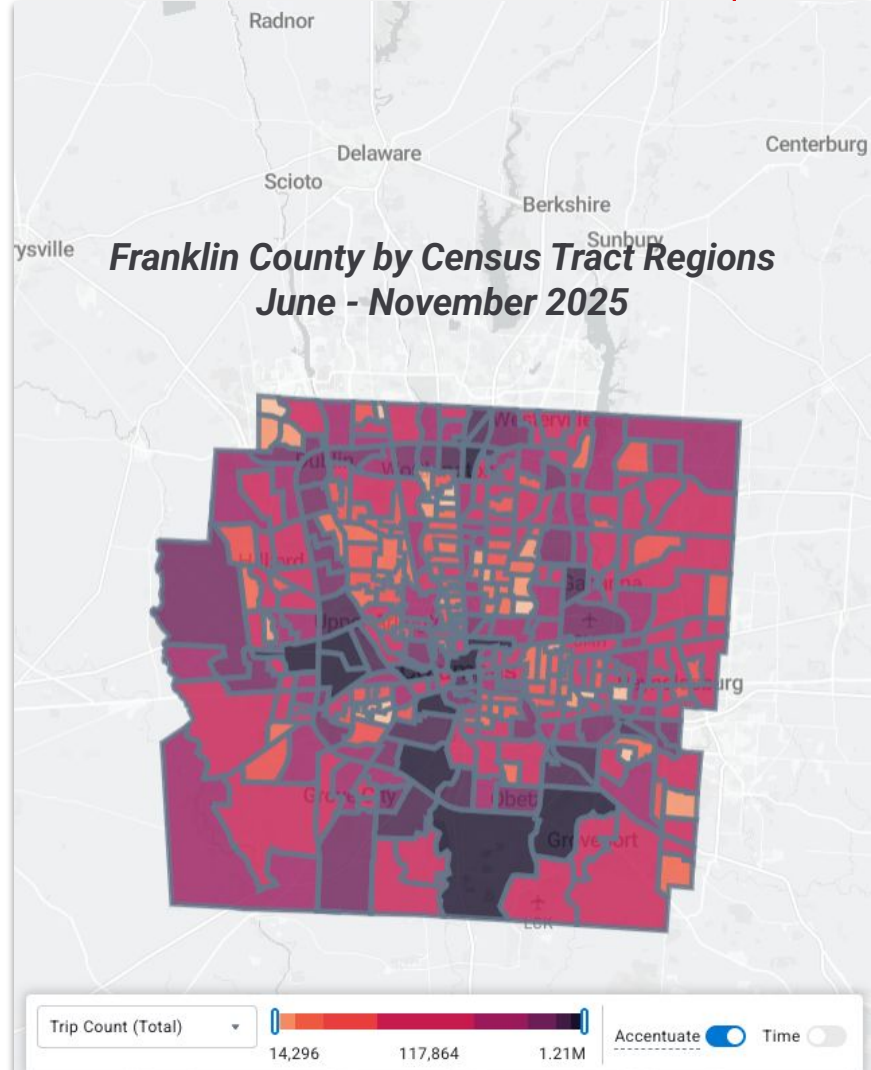
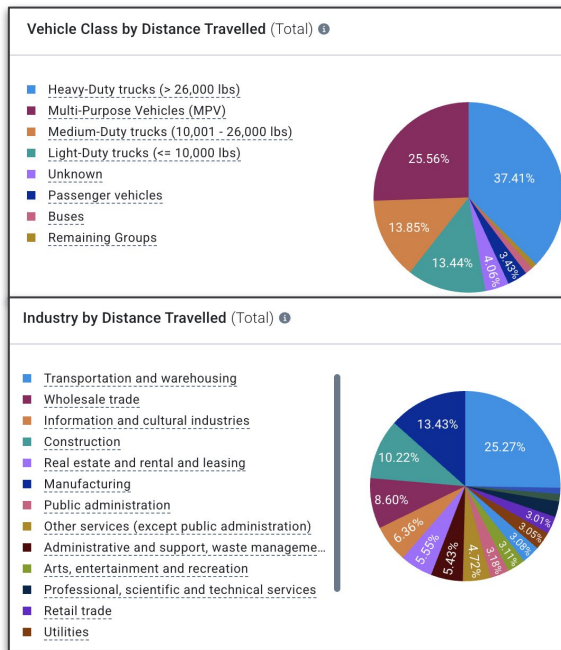


Trip Generation - Aggregated

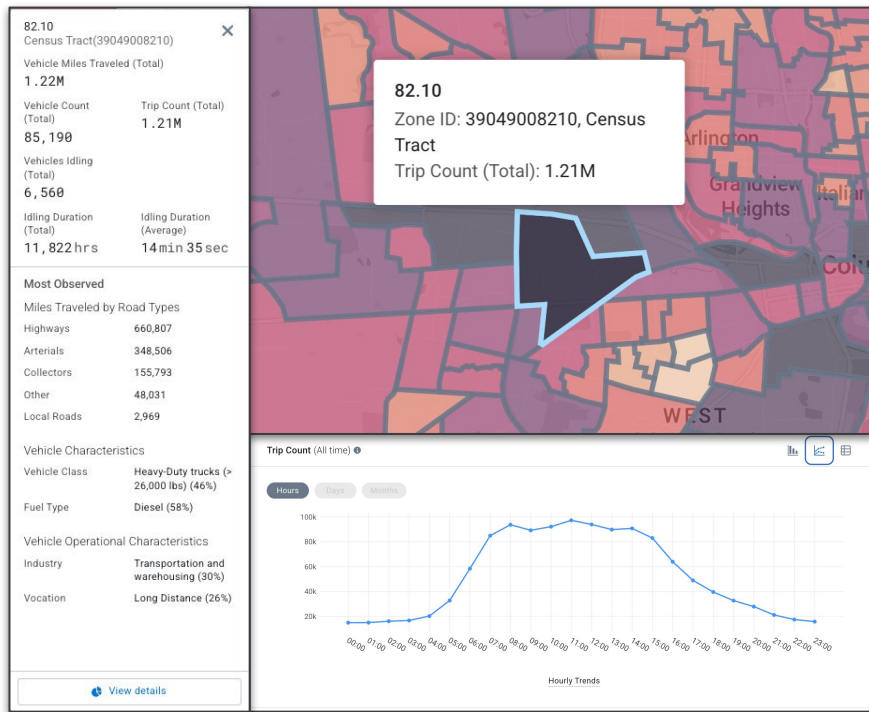
- Trip metrics are further broken down into **vehicle class, industry, vocation, and fuel type.**

Breakdowns also available for other **metrics**:

- Trip counts
- Trip starts
- Trip ends
- Vehicle counts
- Idle metrics



Trip Generation - Individual Zones



- Deep-dive into a selected zone in your analysis to learn more!
- Altitude data can be aggregated across various **temporal** bins (hourly, day of week, monthly) for all or individual zones.

Altitude <> Trip Generation Key Takeaways

Use **Altitude** to augment Trip Generation with commercial vehicle data.

- Altitude can uncover the proportionality of vehicle classes, industries, and vocations.
- Data can be aggregated down to the TAZ-level and hourly-level to enable detailed time and area-specific modeling.

Trip Distribution

Altitude's OD module can be used to determine journey counts from one location to another.

- Compatible with E-E, I-E, E-I, I-I journeys.

Multi-Zone Trip Distribution

- OD analysis can be executed over multiple zones simultaneously.
- Example: I-E analysis from Franklin County to all other counties in Ohio.

OD Analysis - Franklin County to OH Counties
(June - November 2025)

Journey Count: Total ⓘ

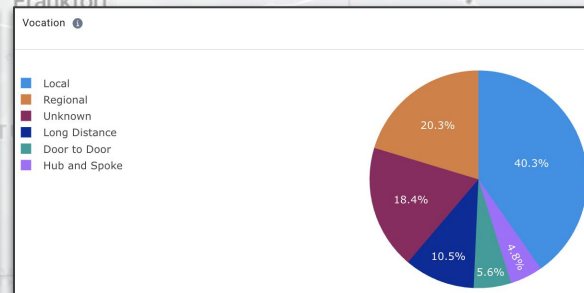
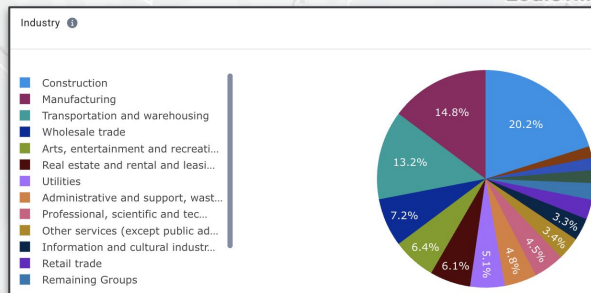
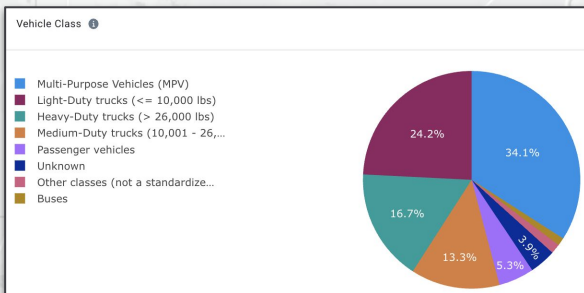
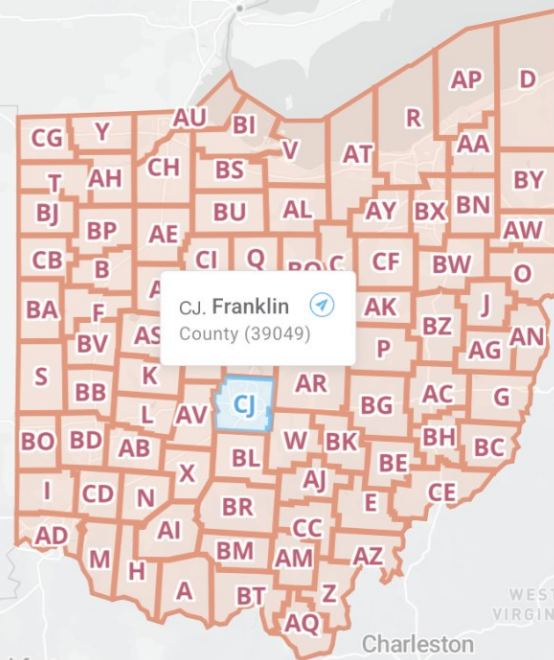
	T. Defiance	U. Delaware	V. Erie	W. Fairfield	X. Fayette
C.J. Franklin	70,518	460	44,791	5,474	
Sum	70,518	460	44,791	5,474	

Distance: Median (miles) ⓘ

	T. Defiance	U. Delaware	V. Erie	W. Fairfield	X. Fayette
C.J. Franklin	22.4	118.7	30.5	46.9	

Journey Duration: Median (minutes)

	T. Defiance	U. Delaware	V. Erie	W. Fairfield	X. Fayette
C.J. Franklin	45	162.8	53.2	69.3	



Freight Generator Distributions

OD Analysis from Distribution Centers near LCK to all OH Counties (June - November 2025)

Journeys Traveling from LCK DC Zone to Counties

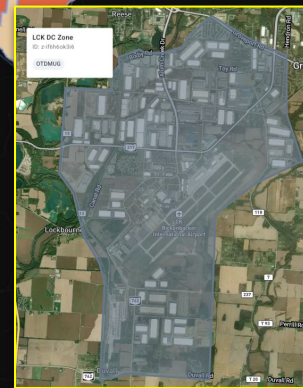
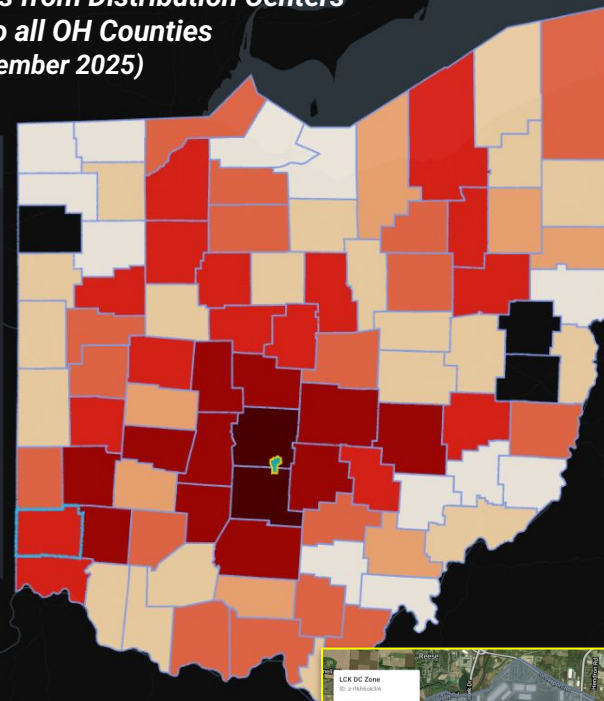


Legend

ClosedOD_LCK_toCounties_Wi

JourneyCount >

- Less than 25
- 25 to 75
- 75 to 100
- 100 to 250
- 250 to 600
- 600 to 15k
- 15k or more



		Franklin	Pickaway	Licking	Fairfield	Madison	Delaware	Montgomery
Median Metrics (minutes)	Journey Duration	28.3	32.7	48.7	38.6	53.8	55.2	118.0
	Journey Distance (miles)	8.5	8.4	30.8	23.7	34.6	37.3	90.0

Stop Analytics

Unlock D2D Delivery Demand

- Understand parking demands
- Observe trends in stop behavior
- Filter by vehicle class, vocation, industry and fuel type
- View in zones, road segments, related roads, and hexagons

Stop Filters [Edit](#) [X](#)

Date range:
10-01-25 - 10-31-25

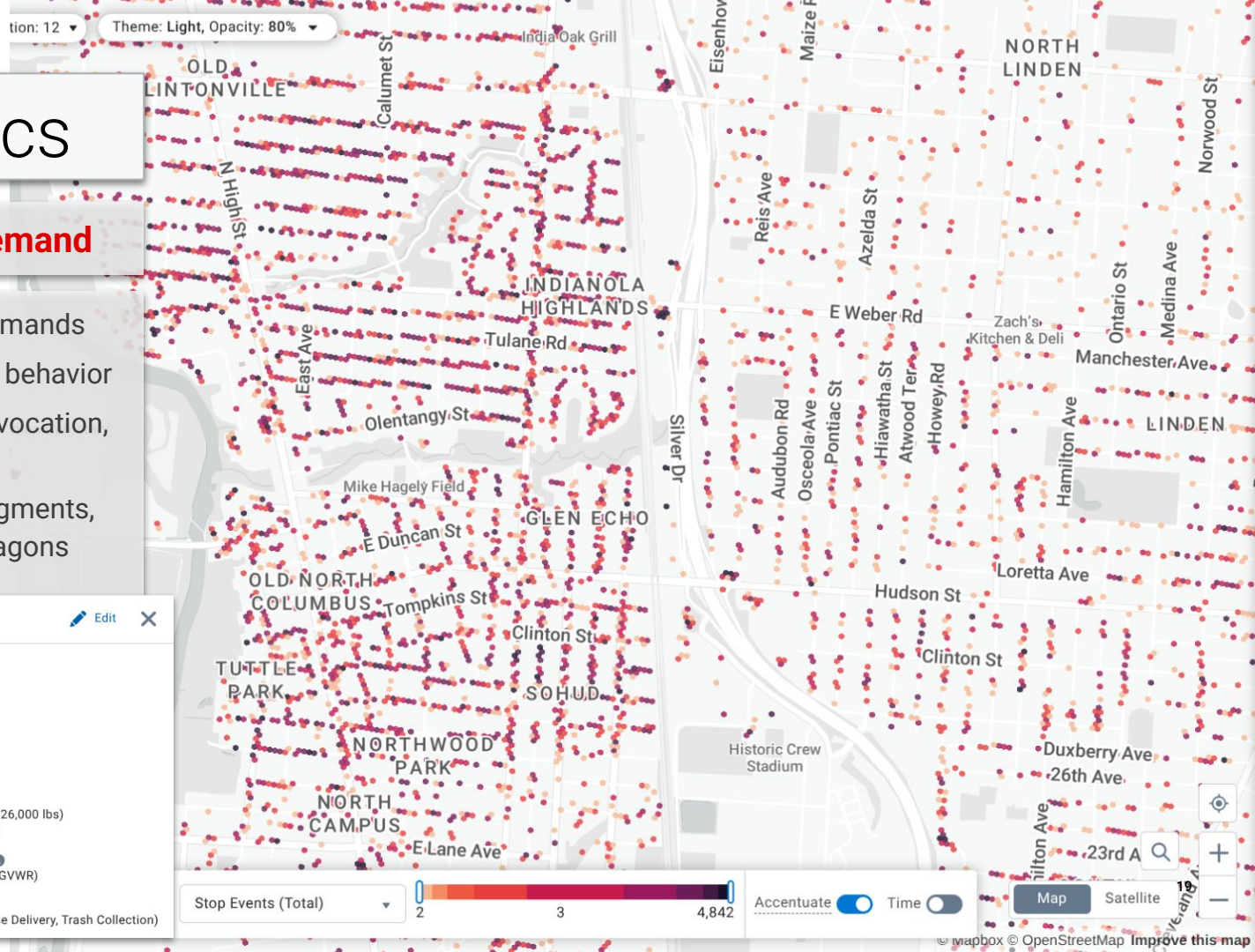
Days of the week:
All valid days

Time of day:
00:00 - 24:00

Vehicle class: [1](#)
Medium-Duty trucks (10,001 - 26,000 lbs)
Multi-Purpose Vehicles (MPV)

Vehicle class truck scheme: [1](#)
Gross Vehicle Weight Rating (GVWR)

Vocation: [1](#)
Door to Door (e.g. Merchandise Delivery, Trash Collection)



Altitude in TDM Trip Distribution

Altitude <> Trip Distribution Key Takeaways

Use **Altitude** to estimate truck volumes and special generator (airport and seaport) volumes for all combinations of internal-external journeys.

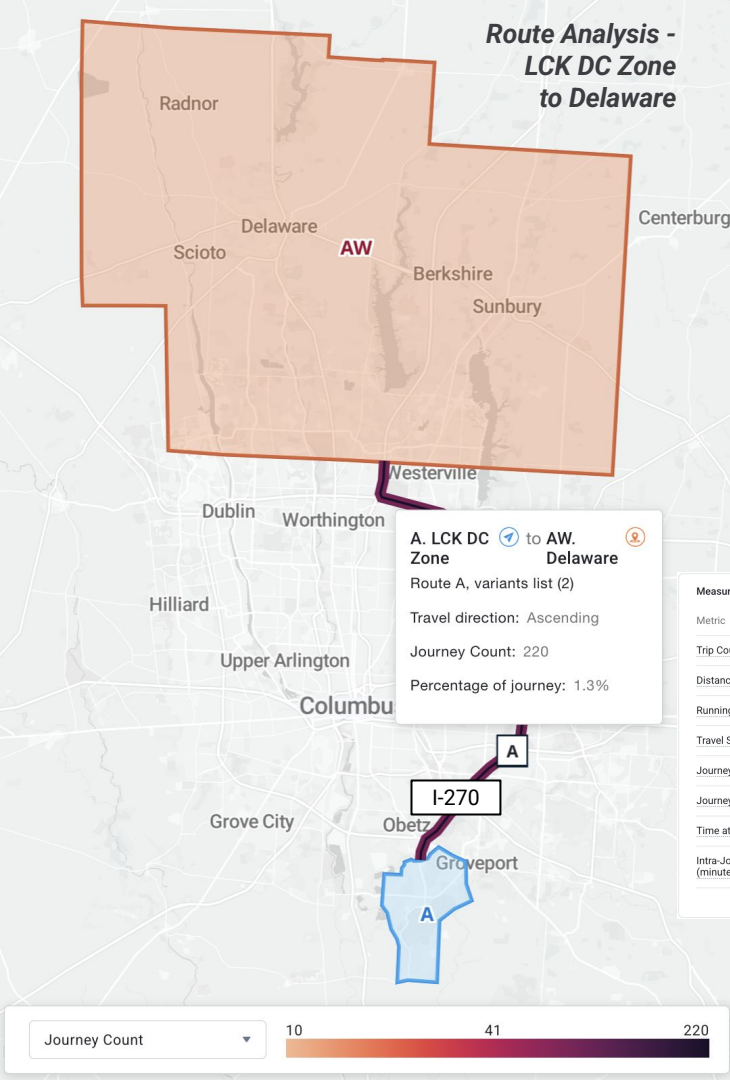
- Altitude can surface the number of trips travelling from internal zones to external & vice versa.
 - **Special Altitude Feature:**
passthrough/connector geographies
- OD analysis can even be executed from freight generators using Custom Zones!

Route Analysis - LCK DC Zone to Delaware

Trip Assignment & Mode Distribution

- Once OD journeys have been established, routes connecting the zones can be analyzed with Altitude.
- We can drill down to each route and determine the mode split of a given route.

Routes (17)		Journey Count	Daily Average	% of Total Journeys	Distance	Running Speed	Travel Speed	Journey Duration	Idle Duration
Q Search road names					Median	Median	Median	Median	Median
A. Alum Creek Drive,Jack Nicklaus FreewayJ 71		220	1.86	1.3%	25.87mi	63.3mph	60.52mph	25.68min	1.41min
B. Lockbourne Road,Rathm ... 71,East InnerbeltJ 71		84	1.38	0.5%	25.29mi	52.56mph	50.07mph	29.85min	0.88min
C. Alum Creek Drive,Jack ... us Freeway,Sunbury Road		60	1.36	0.4%	23.96mi	51.7mph	47.91mph	29.95min	2.33min
D. Alum Creek Drive,Jack ... res),North High Street		53	1.26	0.3%	27.91mi	61.57mph	56.36mph	29.72min	2.35min
E. Alum Creek Drive,Jack ... S 33,SR 161,US 33,US 42		42	1.27	0.2%	37.77mi	56.84mph	53.34mph	42.38min	1.99min
F. London - Groveport Ro ... ay,Olentangy River Road		41	1.05	0.2%	24.56mi	54.55mph	49.29mph	30.3min	2.29min
G. Alum Creek Drive,Jack ... reet,North State Street		34	1.17	0.2%	23.67mi	54.33mph	51.03mph	27.33min	2.05min
H. SR 317,South Hamilton ... res),North High Street		34	1.03	0.2%	27.85mi	56.5mph	54.11mph	30.83min	1.44min
I. Alum Creek Drive,Jack ... 71,East InnerbeltJ 71		33	1.18	0.2%	26.7mi	59.41mph	58.15mph	27.55min	0.53min
J. Alum Creek Drive,Jack ... us Freeway,Sawmill Road		17	1.06	0.1%	29.47mi	55.63mph	53.09mph	33.3min	3.13min



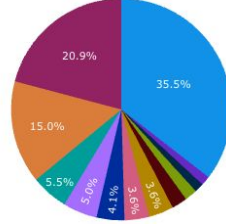
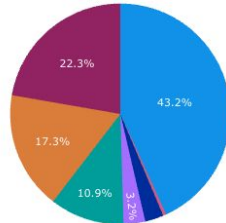
Trip Assignment & Mode Distribution

- Once OD journeys have been established, routes connecting the zones can be analyzed with Altitude.
- We can drill down to each route and determine the mode split of a given route.

Measurements			
Metric	Median	Average	Standard Deviation
Trip Count Per Journey	1	1.9	1.5
Distance (miles)	25.9	25.9	0.1
Running Speed (mph)	63.3	62.4	7.9
Travel Speed (mph)	60.5	58.4	8.1
Journey Duration (minutes)	25.7	27.5	7
Journey Idle Duration (minutes)	1.4	1.7	1.4
Time at Destination (minutes)	105.8	381.7	708.1
Intra-Journey Stop Duration (minutes)	0	0.5	4.4

- Heavy-Duty trucks (> 26,000 lbs)
- Multi-Purpose Vehicles (MPV)
- Medium-Duty trucks (10,001 - 26,000 lbs)
- Light-Duty trucks (<= 10,000 lbs)
- Unknown
- Passenger vehicles
- Other classes (not a standardize...)

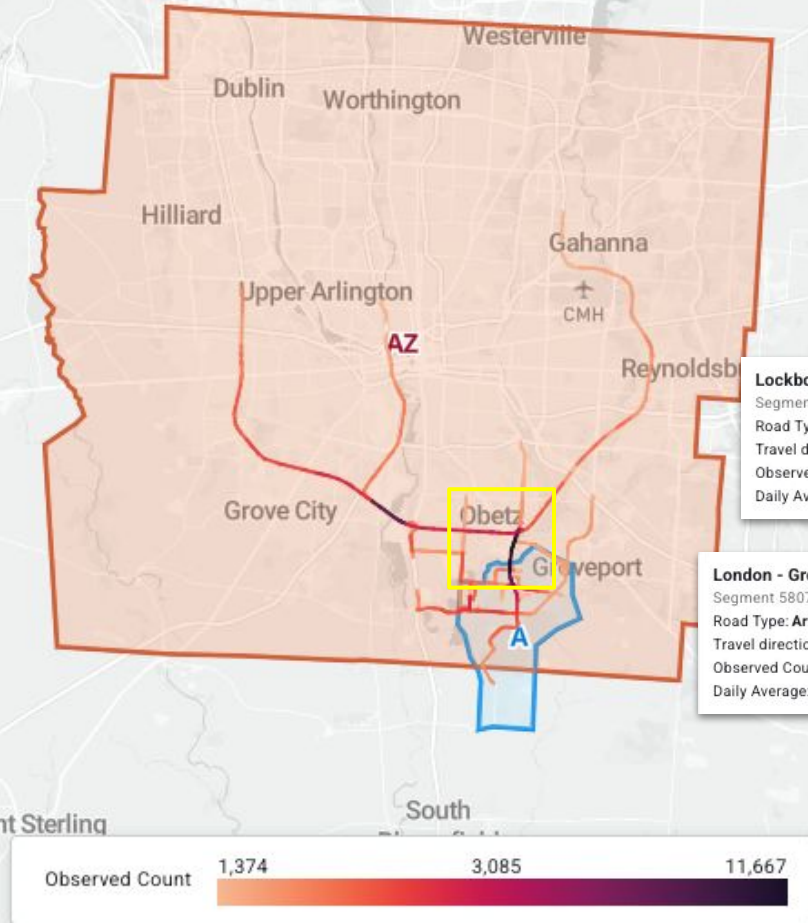
- Manufacturing
- Transportation and warehousing
- Construction
- Information and cultural industr...
- Administrative and support, wast...
- Wholesale trade
- Retail trade
- Professional, scientific and tec...
- Real estate and rental and leasi...
- Arts, entertainment and recreati...
- Other services (except public ad...
- Remaining Groups



Route Analysis - LCK DC Zone to Franklin

Trip Assignment & Mode Distribution

- Altitude also makes analyses on common roads and routes possible on adjacent OD pairs.

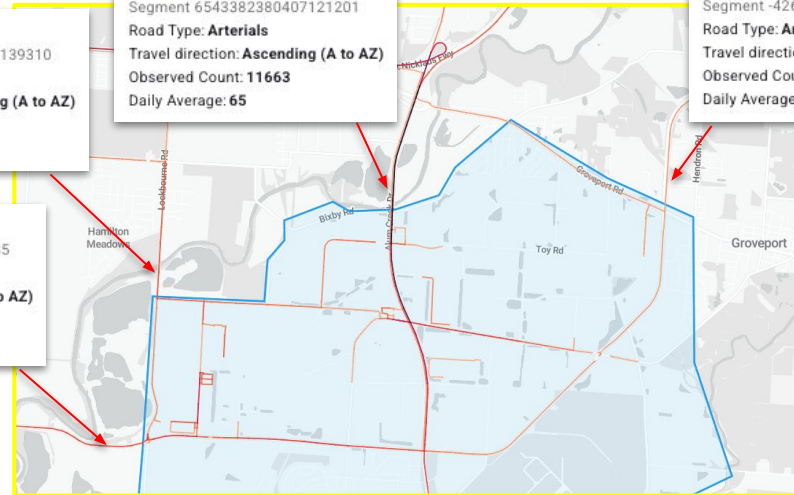


Lockbourne Road
 Segment 5065082538432139310
 Road Type: **Arterials**
 Travel direction: **Ascending (A to AZ)**
 Observed Count: **4184**
 Daily Average: **23**

London - Groveport Road
 Segment 5807153357853323285
 Road Type: **Arterials**
 Travel direction: **Ascending (A to AZ)**
 Observed Count: **4311**
 Daily Average: **24**

Alum Creek Drive
 Segment 6543382380407121201
 Road Type: **Arterials**
 Travel direction: **Ascending (A to AZ)**
 Observed Count: **11663**
 Daily Average: **65**

SR 317
 Segment -4269528587943423768
 Road Type: **Arterials**
 Travel direction: **Ascending (A to AZ)**
 Observed Count: **2198**
 Daily Average: **13**



Altitude in TDM Trip Assignment & Mode Distribution

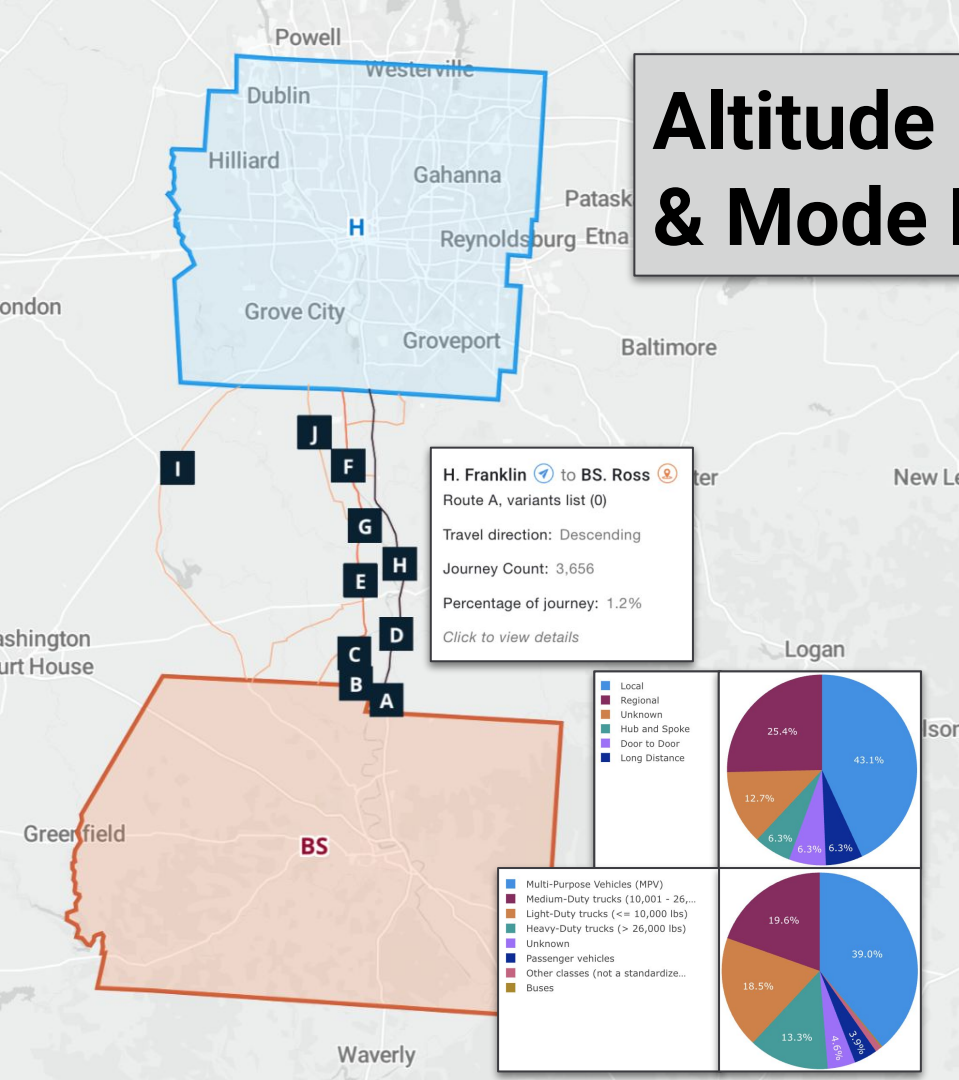
Altitude <> Trip Assignment Key Takeaways

Use **Altitude** to enhance Trip Assignment by unlocking insights into key routes for O-D pairs.

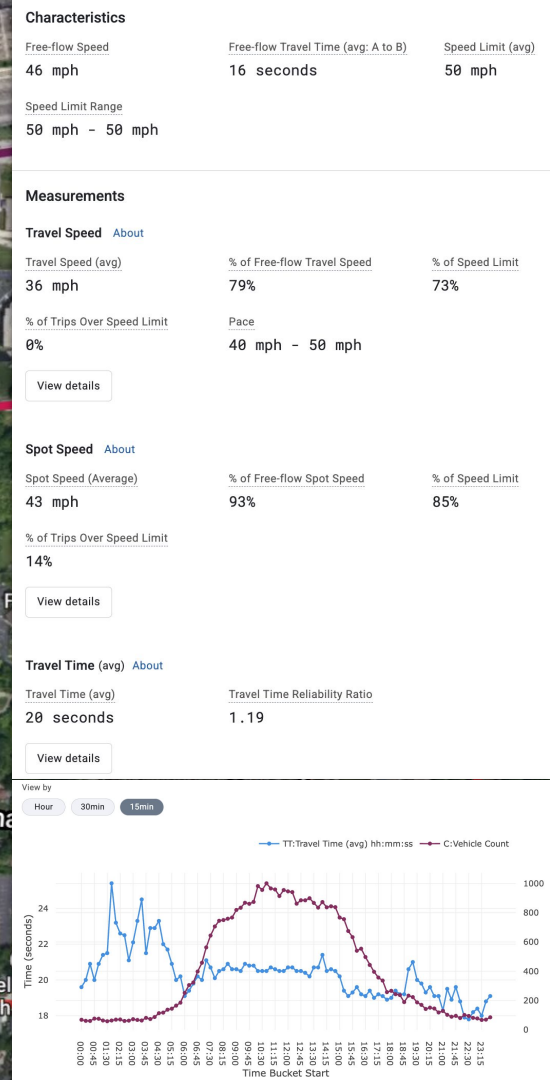
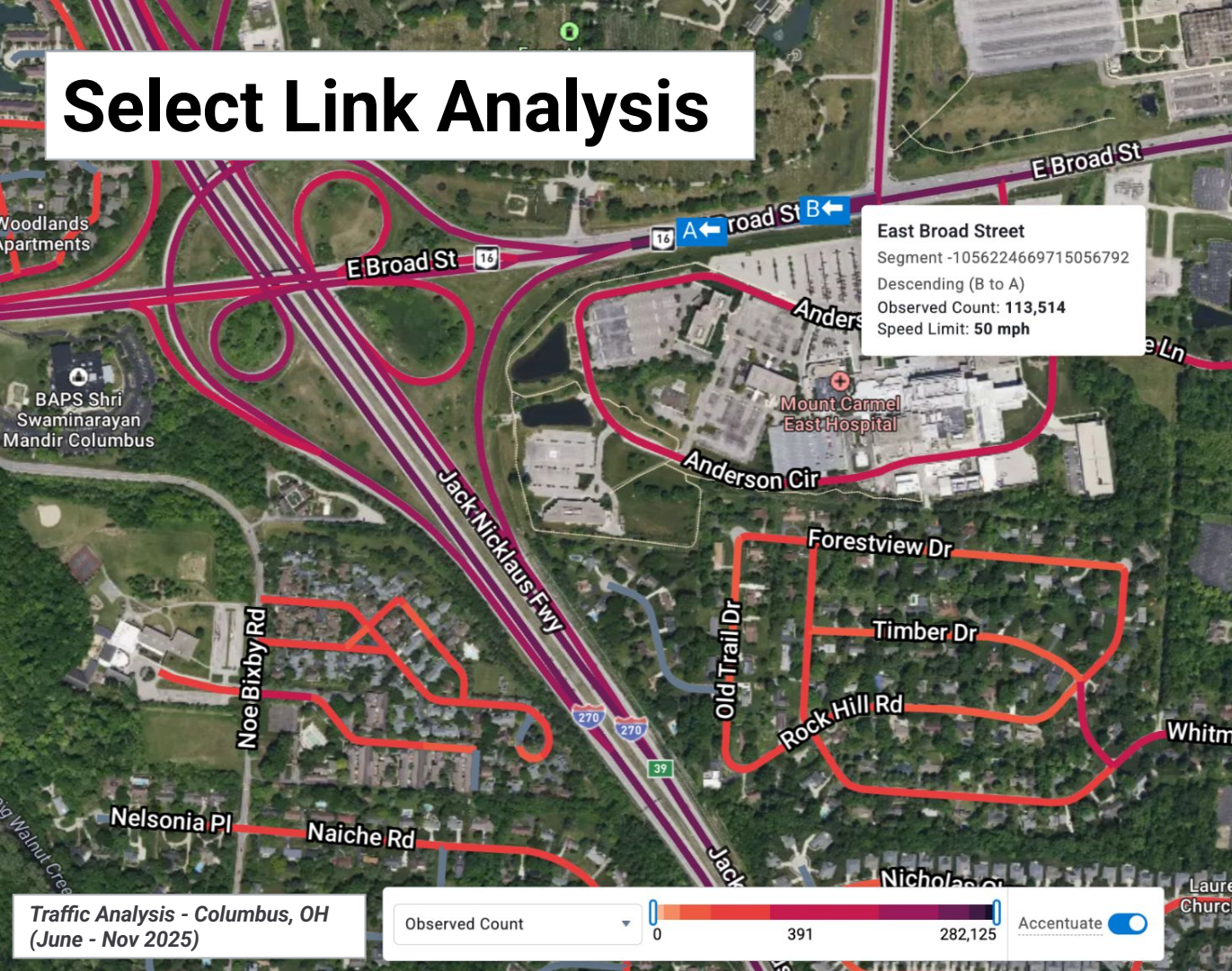
- Travel metrics such as journey counts, duration, and distance can be surfaced for key routes.

Mode Distribution

- Vehicle breakdowns (class, industry, vocation) can provide context of the makeup of traffic flowing through these key routes.



Select Link Analysis



Segment Flow Analysis

Segment Flow Analysis
I-71 Northbound & I-270
 (June - November 2025)

OD using road segments can be leveraged for segment flow analyses.

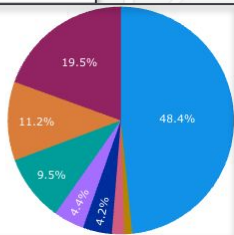
- How much traffic is diverted from I-71 Northbound to I-270?

Journey Count: Percentage (%) ⓘ

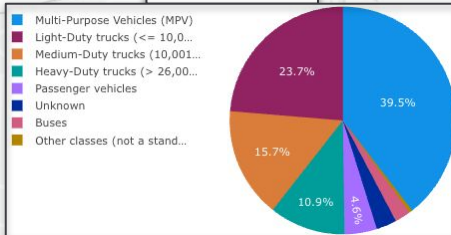
Destinations	B. I-71	C. I-270 EB	D. I-270 WB	Sum
Origins				
A. I-71	34.5	29.2	36.3	100
Sum	34.5	29.2	36.3	100

- What types of vehicles make up each segment flow?
 - Most HDT continue northbound (B, 48.3%), westbound (D, 25.7%), then eastbound (C, 10.9%).

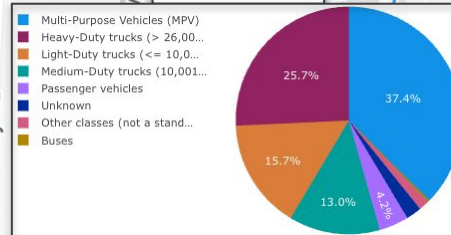
A → B



A → C



A → D



Expansion Factors & AADT

Altitude offers modeled **expansion factors** and **AADT** to enable users to scale metrics beyond Geotab's data.

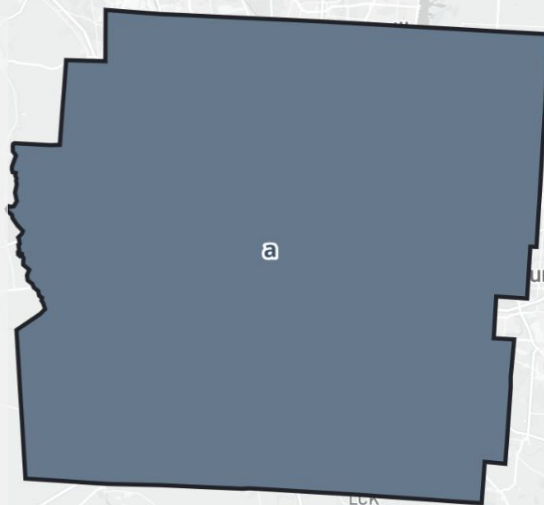
- Trained using ML methods.
- **Key data sources:** **Geotab** vehicle data (**observed**), **FHWA HPMS** data (**ground-truth**), **US census socio-economic** data, **FHWA and S&P** vehicle registrations.

Expansion Factors (EF)

- Enables users to **extrapolate** Geotab's observed data into estimation of **market volumes**.
- Altitude currently offers 2022/23 EF at the **State and County** levels.
- HDT/MDT/Universal @ various road types.
- 2024/25 release expected 2026.

Annual Average Daily Traffic (AADT)

- Altitude currently offers 2022 AADT at the **road segment** level.
- 2023/24/25 release expected H1 2026.



Location Details

County Franklin ID 39049

Country United States

Filters

Vehicle Group: ⓘ

Heavy Duty Medium Duty Universal

Road Type:

All Motorway Primary Secondary

Tertiary Trunk

Expansion Factors

Expansion Factor	CI 95%
34.1x	2.5 +/-
Marketshare est.	Heavy Duty Vehicles
2.9%	34.1x

Error measures

Level of fit ⓘ	MAPE
 High	7.3%

Resources:

[Altitude coverage breakdown – United States](#)
[Expansion Factors: Technical White Paper Version 3.0](#)

Altitude & OTDMUG

How can Altitude help improve Travel Demand Models?

Diversifying Model Data Sources

Altitude's position as a leading source of **commercial** data helps fill gaps in data.

- Freight generators.

Breakdowns by **industry**, **vehicle class**, and **vocation** provide context into vehicle distributions for trips and traffic.

Data is continuously being collected - future opportunities for model year updates (**stability**).

Enhancing Travel Demand Models

Altitude's trip counts, starts, and ends augment the **Trip Generation** stage of the model.

The O-D and route analysis modules help determine E-I, E-E, I-I routes for **Trip Distribution** and **Trip Assignment**.

Traffic analytics allows for insight into **Select Link Analysis** (travel times, speeds, volumes) at the road segment level.

Compatibility and Validation

Altitude can provide data in various temporal formats (**hourly**), and can perform analysis down to the road segment level (segment flows).

Altitude's data can be used as a **validation** tool against model outputs.

In addition to modules, Altitude provides **AADT** and **Expansion Factors** to estimate expanded volumes (based upon FHWA counting stations).

Elevate your transportation plans.



Christopher Wichman

Business Development
Manager

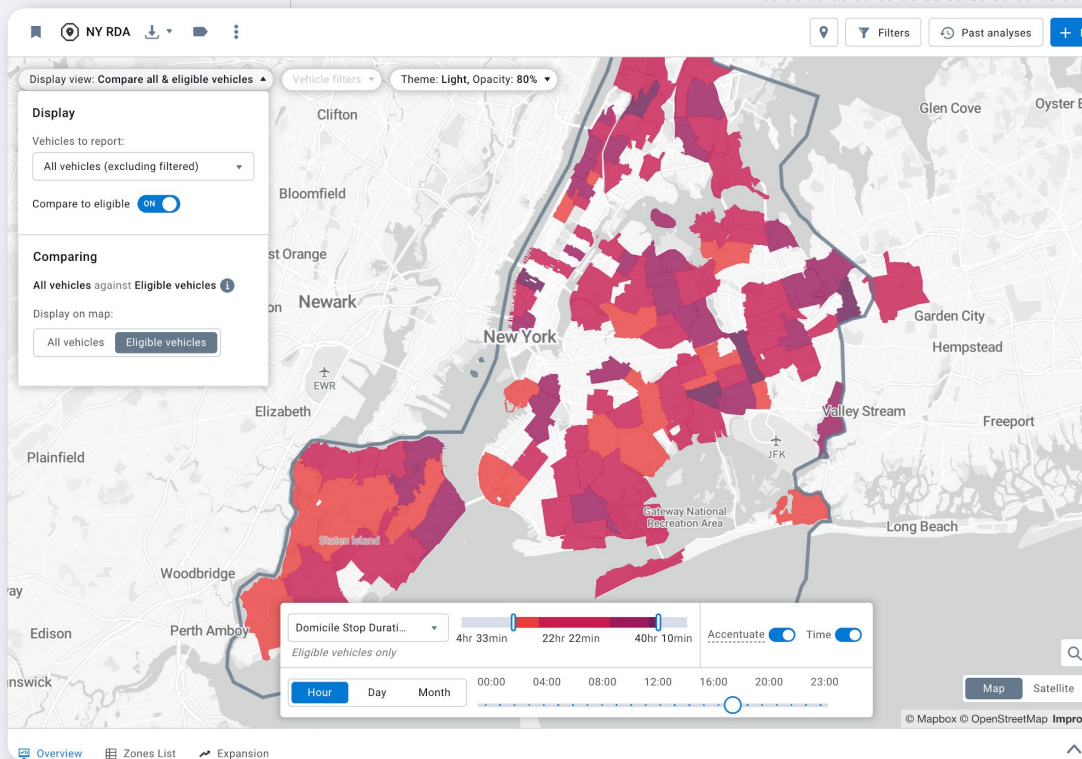
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AltitudeTM
by GEOTAB

